

DRAWVLA-VERIFY: VALIDATING SKETCHED ROBOT INSTRUCTIONS BEFORE EXECUTION

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ABSTRACT

Sketches let a person tell a robot *which* object to manipulate (a circle) and *where* it should go (an arrow) without naming anything, but existing sketch-conditioned robot systems assume the drawing is correct. We propose **sketch-prompt validation**—deciding *before* any action whether a drawn instruction is correct, consistent across circle, arrow, and caption, and sufficient for a unique action, and saying why not—as an explicit, benchmarked task. A pilot simulator-grounded generator over LIBERO yields 37,970 labelled prompts; frontier vision-language models (VLMs) fail it zero-shot (best 0.79 accuracy), a 500M LoRA-tuned validator reaches 0.90, yet a geometry-only classifier reaches 0.93, exposing shortcuts. We propose to (H1) confirm learnability against strong baselines, (H2) remove shortcuts with a distribution-matched, adversarially filtered generator audited by scene-blind ceilings, (H3) *generate* a per-predicate verdict and rejection reason behind a conformal false-accept guarantee, and (H4) test transfer to human sketches.

1 PROBLEM AND MOTIVATION

Language is ambiguous in cluttered scenes with duplicate objects and goal images are over-specified (Sundaresan et al., 2024); a circle-and-arrow sketch with a name-free caption (“move this there”) is a persistent deictic act (Saupé & Mutlu, 2014; Kranstedt et al., 2006). Recent systems condition robot policies on such sketches (Sundaresan et al., 2024; Gu et al., 2024; Li et al., 2025a;b) and all silently assume the drawing is *valid*. Yet circles miss or enclose a distractor, arrows point to the wrong container or detach from the circle, captions contradict sketches, and channels that look fine alone can jointly disagree; executing such a sketch commits the robot to the wrong action.

Problem (Track B). *Can a model decide, before any action, whether a drawn instruction is safe to execute—and say why not?* We formulate **sketch-prompt validation** as a decision over image, circle, arrow, and caption requiring per-channel correctness, cross-channel consistency, and sufficiency

*CS707 Group Project Proposal, Track B (novel task formulation). Replace placeholder names and e-mails before submission.

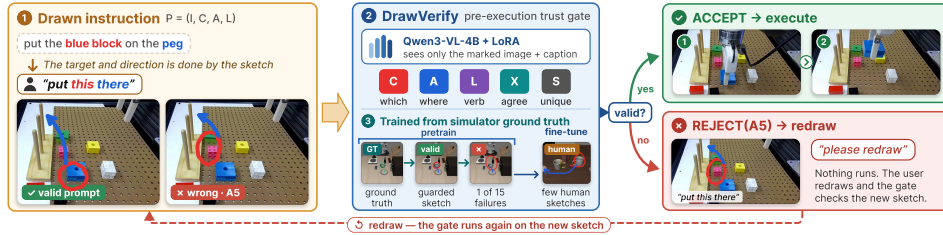


Figure 1: The pilot validator as a pre-execution trust gate (DrawVLA-Verify pilot, 2026): a LoRA-adapted VLM trained on simulator ground truth judges the marked image and caption; ACCEPT executes, REJECT asks the user to redraw.

(Section 3.1), plus a generated reason; no existing task captures it (Section 2). We address it with generative VLMs as zero-shot judges and as LoRA-tuned validators (Figure 1).

Pilot evidence and gap. A pilot by team members (DrawVLA-Verify pilot, 2026) built DrawVLA-Verify, a simulator-grounded generator over LIBERO (Liu et al., 2023a) producing 37,970 labelled prompts across 40 tasks with no manual annotation. Frontier VLMs fail zero-shot (best 0.79 accuracy, 0.43 MCC; majority class 0.74) while SmolVLM-500M (Marafioti et al., 2025) with LoRA (Hu et al., 2022) reaches 0.90 (Table 3, Appendix B); but a geometry-only classifier reaches 0.93, i.e., the generator leaks shortcuts (Geirhos et al., 2020), and the pilot was single-seed with one zero-shot prompt and no ablations, human sketches, or explanations. This project delivers those deltas.

Hypotheses. **H1 (learnability):** a $\leq 1\text{B}$ LoRA-tuned VLM beats the best zero-shot VLM under its best prompt by ≥ 10 accuracy points and is within 1 balanced-accuracy point of, or exceeds, a fine-tuned discriminative model of similar size, over 3 seeds. **H2 (shortcut removal):** on the hardened tier the multimodal validator keeps ≥ 0.80 balanced accuracy and exceeds both scene-blind ceilings (geometry-only and blank-canvas, on the untouched audit split) by ≥ 15 points. **H3 (structured supervision):** supervising the five validity predicates as auxiliary generative targets improves the directional and grounding slices (the pilot’s weakest) by ≥ 3 points over binary-only training, and generated reasons name the violated predicate with $\geq 80\%$ faithfulness. **H4 (sim-to-human transfer):** validators trained with guarded imprecision augmentation lose ≤ 10 balanced-accuracy points on human sketches; removing it widens the gap by ≥ 5 points. **Gate:** empirical test FAR at $\hat{\tau}$ (Eq. 2) is ≤ 0.05 in every seed on v1 and v2.

2 RELATED WORK

Sketch- and visual-prompt-based instruction. RT-Sketch and RT-Trajectory condition policies on goal or trajectory sketches (Sundaresan et al., 2024; Gu et al., 2024), RoVI decodes an arrow/circle language with a VLM (Li et al., 2025b), CrayonRobo *tolerates* noisy prompts but does not classify them (Li et al., 2025a), Action-Sketcher and AnyUser generate or translate sketches (Tan et al., 2026; Yang et al., 2026), and visual-prompting methods (Nasiriany et al., 2024; Yang et al., 2023; Liu et al., 2024; Yuan et al., 2024; Cai et al., 2024) annotate images so a VLM can ground spatial decisions. All assume the prompt is valid; *we extend* them by making validity the primary decision.

Verification and safety gating. KnowNo asks for help via conformal prediction (Ren et al., 2023); VerifyLLM and VeriGraph verify plans against temporal logic or scene graphs (Sautenkov et al., 2025; Ekpo et al., 2024); DoReMi and REFLECT detect or explain execution failures (Guo et al., 2024; Liu et al., 2023b). These verify *plans* or *executions*; *we contrast* by verifying the instruction itself, an earlier gate, with a conformal threshold (Angelopoulos et al., 2024; Geifman & El-Yaniv, 2017). **Shortcuts.** A geometry-only classifier is a hypothesis-only baseline in the NLI sense (Gururangan et al., 2018); AFLite (Le Bras et al., 2020) and dataset cartography (Swayamdipta et al., 2020) remove or expose artefacts (Geirhos et al., 2020) in fixed datasets; *we control the generator*.

3 TASK FORMULATION AND METHOD SKETCH

3.1 FORMULATION

A sketch prompt is $P = (I, C, A, L)$: RGB observation I , circle C (centre, radius) denoting *which* object, arrow A (tail, head) denoting *where* it goes, and a name-free caption L . From simulator ground truth we define predicates v_C (the circle uniquely and correctly selects the target), v_A (correct bearing and magnitude, anchored on the circled object, ending in the intended region), v_L (caption verb and implicit referents match the intended action), cross-modal consistency v_X (all channels describe the same object, destination, and predicate), and sufficiency v_S (the prompt determines a unique executable action). A prompt is valid iff $V(P) = v_C(P) \wedge v_A(P) \wedge v_L(P) \wedge v_X(P) \wedge v_S(P)$; the task is to learn $f_\theta(P) \in \{\text{ACCEPT}, \text{REJECT}\}$ predicting $V(P)$, with a reason on REJECT. A prompt can pass every per-channel predicate yet be invalid because $v_X = 0$ or $v_S = 0$, so validation is not a set of independent checks.

3.2 SIMULATOR-GROUNDED DATA GENERATION AND SHORTCUT HARDENING

Pilot pipeline (Figure 1; Appendix Figure 3, top). From LIBERO’s object poses, region geometry, and gripper trajectories we extract the correct circle target and arrow endpoints for every frame with no human labelling. Valid prompts get *guarded* human-imprecision augmentation (jitter, non-circular strokes, hand-like shafts): an in-point guard keeps the target inside the circle, an out-point guard keeps distractors outside. Negatives apply one controlled corruption from four families (14 of 18 modes implemented; Appendix A) plus five grounding configurations, so every negative has a known mode, severity, and violated predicate.

New: shortcut hardening (H2). Let $g(P)$ be the raw geometry vector (circle centre and radius, arrow endpoints, length, bearing). (i) *Distribution matching*: negatives draw the same imprecision magnitudes as positives on the valid side of a guard, and corruptions that must cross a guard (near-miss, wrong magnitude) draw severities from the mirrored distribution just beyond it, so class-conditional marginals of g overlap as far as labels allow. (ii) *Adversarial filtering* (Le Bras et al., 2020) on medium/hard modes only: we repeatedly train geometry-only gradient-boosted classifiers on random folds, score each example by how often held-out classifiers predict it confidently (> 0.75), and remove the most predictable until geometry-only balanced accuracy on an *untouched* audit split (also probed with MLP and k -NN) is ≤ 0.60 or retention falls below 60%. Since pixel artefacts can leak even when coordinates do not, we also report a *scene-blind* ceiling (sketches on a blank canvas).

3.3 STRUCTURED VALIDATOR AND CONFORMAL INFERENCE

Model. A compact VLM (SmoVLM-256M/500M) with LoRA takes the image with the rendered sketch, the caption, and a fixed prompt and generates a *structured verdict*: five predicate questions q_k , $k \in \{C, A, L, X, S\}$, answered sequentially with earlier answers in context (teacher-forced in training), then the ACCEPT/REJECT question q_V , then, on REJECT, a one-sentence reason. Predicate labels are free (each negative violates a known subset; positives satisfy all); we also report the derived decision $\hat{V}' = \bigwedge_k \hat{v}_k$ and its disagreement with $\hat{V}$.

Objective. Each yes/no answer is scored by forced choice over its two answer tokens, $p_\theta(\text{yes} \mid P, q_k) = \sigma(z_{\text{yes}} - z_{\text{no}})$; the reason uses token-level cross-entropy ℓ_R on REJECT examples against one of ≥ 3 paraphrased template reasons per mode (one held out). We minimise

$$\mathcal{L}(\theta) = \ell_V(\theta) + \lambda \sum_{k \in \{C, A, L, X, S\}} \ell_k(\theta) + \mu \ell_R(\theta), \quad \ell_k(\theta) = -\log p_\theta(y_k \mid P, q_k), \quad (1)$$

with λ, μ tuned on validation; $\lambda = \mu = 0$ recovers the pilot’s binary-only validator (ablation A1).

Inference with a finite-sample false-accept guarantee. Let $s(P) = p_\theta(\text{ACCEPT} \mid P, q_V)$; we ACCEPT iff $s(P) \geq \hat{\tau}$. False accepts (executing an invalid prompt) are the safety-critical error, so we bound the false-accept rate (FAR) by conformal risk control (Angelopoulos et al., 2024): on the n *invalid* validation prompts let $\ell_i(\tau) = \mathbb{I}[s(P_i) \geq \tau]$ and

$$\hat{\tau} = \inf \left\{ \tau : \frac{1}{n+1} \sum_{i=1}^n \ell_i(\tau) + \frac{1}{n+1} \leq \alpha \right\}. \quad (2)$$

If invalid validation and test prompts are exchangeable, $\mathbb{E}[\text{FAR}(\hat{\tau})] \leq \alpha$ ($\alpha = 0.05$); we report the induced false-reject rate and empirical coverage across seeds, tiers, and the human set (where exchangeability is *not* guaranteed); rejected prompts return the reason.

4 DATASETS, METRICS, BASELINES, AND EXPERIMENTAL PLAN

Datasets and licences. (1) *LIBERO* (Liu et al., 2023a) (MIT): 800 demonstrations, 40 tasks. (2) *DrawVLA-Verify v1* (pilot): 37,970 examples—30,302 train / 3,702 val / 3,966 test, demo-disjoint; 9,098 positives, 28,872 negatives (Appendix A). (3) *v2 (hardened)*: same splits after matching and filtering ($\geq 60\%$ retention expected) plus the untouched audit split. (4) *Human sketch test set* (evaluation only; cf. Figure 2; SMU IRB exemption sought first): ≈ 10 adult volunteers each draw circle+arrow prompts on 30 demo-disjoint frames, 15 *unprompted* and 15 *deliberately wrong* (≈ 300 sketches; no captions, so v_L/v_X are out of scope); guards give labels, two members annotate blind. Data and adapters: CC BY 4.0; code: MIT.

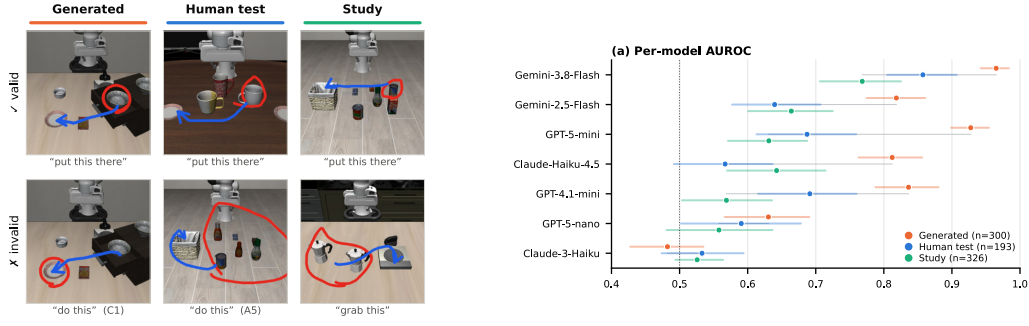


Figure 2: Pilot evidence for the evaluation plan (DrawVLA-Verify pilot, 2026). *Left*: three evaluation sources, one valid (top) and one invalid (bottom) prompt each: generated prompts with a known failure mode (C1: circle on the plate, arrow from the bowl), held-out human sketches (A5: one circle encloses several bottles), and in-the-loop study sketches (both moka pots circled, so “grab this” has no unique referent). *Right*: AUROC of seven API validators on the three sources (95% bootstrap intervals): generated prompts rank models like human sketches do but inflate absolute scores unevenly, motivating H2 and H4.

Metrics. Accuracy, balanced accuracy, macro-F1, MCC (74% negative test set); FAR and FRR at $\hat{\tau}$ and FAR at FRR = 10%; per-family and per-mode slice accuracy; geometry-only and scene-blind ceilings per tier; the transfer gap Δ_{human} (balanced accuracy); reason faithfulness on true rejects against the ground-truth predicate and the model’s own $\hat{v}$ (string match and a Qwen3-8B judge); a 5-point usefulness rating of 50 reasons.

Baselines. (i) Majority class. (ii) Zero-shot open VLMs Qwen3-VL-2B/4B/8B (Qwen Team, 2025) and (iii) closed APIs (Gemini 3.7 Flash, Claude Haiku 4.5/Sonnet 4.6), pinned and cached; per model the best of direct, chain-of-thought, 4-shot, and coordinates-as-text prompts is chosen on a 500-example validation subset. (iv) Geometry-only gradient boosting (Ke et al., 2017) and the scene-blind validator as ceilings. (v) A fine-tuned discriminative model of similar size (ConvNeXt-T with late fusion of $g(P)$ and a caption embedding). (vi) The pilot’s binary-only SmolVLM+LoRA. *Ours*: structured SmolVLM+LoRA with the conformal gate.

Ablations (≥ 3), sensitivity, statistics. A1: structured (+reason) vs. binary-only ($\lambda = \mu = 0$; H3). A2: train on v2 vs. v1, test on both tiers (H2). A3: modality (no caption; coordinates-as-text; scene-blind). A4: no guarded-imprecision augmentation (H4). A5: prompt-only vs. LoRA. A6: backbone 256M/500M/2.2B. Sensitivity: training-set size, LoRA rank, τ sweep, corruption severity. All trained models use 3 seeds (mean $\pm$ std); McNemar’s test (Dietterich, 1998) on seed-matched predictions, paired-bootstrap 95% CIs, minimum detectable effects per slice.

5 RISKS, RESOURCES, AND TEAM PLAN

Risks and mitigations. Hardening over-removes or still leaks: per-mode retention curves, removal capped at 40%, both tiers released. **Small human set; guard label noise:** evaluation-only probe with bootstrap CIs; blind audit of 200 positives and 200 negatives, flagging modes with agreement < 0.8 . **API drift, cost, compute:** pinned versions, cached pilot outputs, Qwen3-VL primary, ablations on the 256M model; ≈ 160 GPU-h on one A100-40GB/RTX 4090 plus $\approx$ USD 100–150 of API calls (reproducibility statement). **Schedule:** go/no-go 25 Oct—if the v2 ceiling is still ≥ 0.75 at 60% retention we report v1 plus the best partial tier and shift effort to H3/H4. **Over-claiming safety:** the gate reduces, not eliminates, unsafe executions; no closed-loop control is evaluated.

Roles and timeline (2026). *Member 1 (data):* generator v2, hardening, audit, release. *Member 2 (modelling):* structured objective, reasons, gate, LoRA training, ablations. *Member 3 (evaluation):* API baselines, statistics, human study, ethics. *1–11 Oct:* IRB, recruitment. *12–25 Oct:* generator v2, 3-seed re-run with stronger baselines (H1), validator and gate, A1–A3; go/no-go. *26 Oct–8 Nov:* human study (H4), A4–A6, sensitivity, statistics. *9–12 Nov:* final presentation ($\geq 90\%$ of results); *13–27 Nov:* final report and artifacts.

ETHICS STATEMENT

All training data are simulator renders of tabletop scenes with no people or personal information. The human sketch study involves adult volunteers who give informed consent; only drawings and anonymous ratings are stored. The validator is a trust gate for robot instructions: a false accept could cause an incorrect physical action, so we report false-accept rates with confidence intervals and recommend deployment with human confirmation. We comply with the MIT licence of LIBERO, the Apache-2.0 licences of SmolVLM and Qwen3-VL, and the terms of service of the closed APIs, and release our data under CC BY 4.0 and code under MIT.

REPRODUCIBILITY STATEMENT

We will release the generator, training and evaluation code, `configs/*.yaml`, seeds, an `environment.yml`, a one-click `scripts/run.sh`, cached API outputs, LoRA adapters (<2GB), and both dataset tiers with checksums, together with GPU model, GPU-hours, and peak memory. Compute budget: one NVIDIA A100-40GB or RTX 4090-24GB; ≈ 3 GPU-h per SmolVLM-500M LoRA run (3 epochs, batch 64, bf16, peak memory <20 GB), ≈ 1.5 h per 256M, ≈ 8 h per 2.2B; ≈ 45 runs ≈ 160 GPU-h plus ≈ 10 GPU-h of inference; APIs ($\approx 4k$ images $\times$ 3 models $\times$ 4 prompts; pilot outputs reused) $\approx$ USD 100–150. Pilot training details and the zero-shot prompt are in Appendix B.

AI USE STATEMENT

[To be completed by the team before submission, in accordance with the CS707 academic-integrity policy.]

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A FAILURE TAXONOMY AND DATASET COMPOSITION

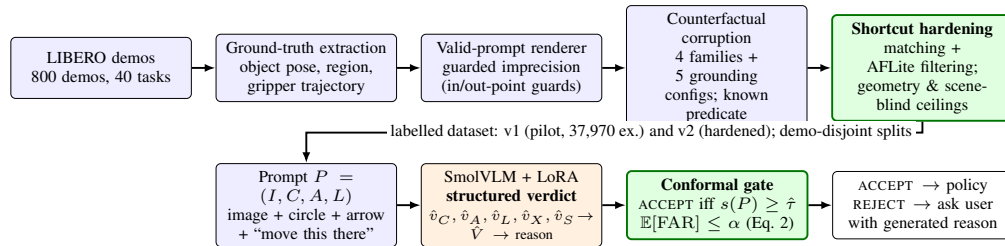


Figure 3: Method sketch (referenced from Section 3.2). *Top*: simulator-grounded data generation (green = new). *Bottom*: the validator generates a structured verdict and a reason; a conformal threshold yields an execute/ask decision.

Table 1: Failure taxonomy. Difficulty: E(asy)/M(edium)/H(ard). Each mode has a controlled generation mechanism and maps to the predicate(s) it violates. Of the 18 corruption modes listed, 14 are implemented in the pilot release; the remaining four are scheduled for v2.

Family	Mode	Description and generation mechanism	Diff.
Referential (circle / “which”; v_C)	Wrong-category object	Circle a clearly different object (e.g., plate, stove) than the target.	E
	Near-miss circle	Offset the centre so the target falls outside or only grazes the circle; severity varies continuously with offset.	H
	Empty-region circle	Circle bare table or background containing no object.	E
	Multi-object enclosure	Inflate the radius to enclose the target <i>and</i> a distractor, making the referent unresolvable (tests v_S).	H
	Degenerate size	Dot-sized or whole-frame circle that selects nothing meaningful.	E
Directional (arrow / “where”; v_A)	Wrong destination	Retarget the arrow head to a different container or region.	H
	Wrong magnitude	Correct bearing but arrow far too short or overshooting.	H
	Wrong bearing	Correct length but head rotated $90^\circ/180^\circ$.	M
	Null destination	Head on empty table or off-frame.	E
	Detached tail	Shaft plausible but tail not anchored on the circled object.	H
	Spec mismatch	Arrow present on a circle-only task, or missing on a place task.	H
Cross-modal (channels disagree; v_X)	Circle–arrow mismatch	Circle object A while the arrow originates from object B.	H
	Caption–sketch mismatch	Correct sketch, but caption swapped from another demo/task.	H
	Predicate mismatch	Caption says “open the cabinet” while the sketch is a place-style circle+arrow.	M
Language (caption / “what”; v_L)	Wrong target noun	Caption names the wrong object.	M
	Wrong destination noun	Caption names the wrong destination.	E
	Wrong verb	Caption verb wrong (pick vs. push vs. open).	M
	Under-determined	“Put this there”: valid with a good sketch, invalid with a bad one.	H
Grounding (v_S)	Five configurations	Fully resolved prompt; no sketch; open destination; identical referents; arrow between destinations.	M–H
Physical (deferred)	Occluded target	Circle a target not visible in the frame (needs depth); generator yielded only 18 rows in the pilot.	M

Table 2: DrawVLA-Verify v1 composition by family (train / validation / test).

Family	Train	Val	Test
Referential	6,354	766	834
Directional	7,338	904	958
Cross-modal	1,212	150	158
Language	2,556	308	336
Grounding	5,648	695	655
Positive	7,194	879	1,025
Total	30,302	3,702	3,966

B PILOT RESULTS, TRAINING DETAILS, AND ZERO-SHOT PROTOCOL

Table 3: Pilot results on the 3,966-example v1 test set (single seed; zero-shot rows use one direct prompt; the test set is 74% negative). [†]Shortcut ceiling from raw coordinates only.

Model	Acc	BAcc	Macro-F1	MCC
Majority class (always REJECT)	0.74	0.50	0.43	0.00
Qwen3-VL-2B (zero-shot, open)	0.32	0.54	0.30	0.13
Qwen3-VL-4B (zero-shot, open)	0.46	0.61	0.46	0.22
Claude Sonnet 4.6 (zero-shot, API)	0.67	0.59	0.59	0.18
Claude Haiku 4.5 (zero-shot, API)	0.73	0.50	0.44	0.01
Gemini 3.7 Flash (zero-shot, API)	0.79	0.71	0.71	0.43
SmolVLM-256M + LoRA (binary-only)	0.89	0.83	0.84	0.69
SmolVLM-500M + LoRA (binary-only)	0.90	0.84	0.86	0.72
Geometry-only HGB [†]	0.93	0.91	0.91	–

Fine-tuning. SmolVLM-256M and -500M with LoRA rank 16, $\alpha = 32$, dropout 0.05, 3 epochs, batch size 64, learning rate 10^{-4} with cosine schedule and 3% warm-up. Decisions are read by forced-choice scoring over the RIGHT/WRONG answer tokens rather than free-form generation. The pilot adapters were trained on an earlier 34,018-row release that contained 14 physical-family negatives; all reported metrics are computed on the identical 3,966-row physical-free test set, and the proposed runs will retrain on the physical-free data.

Zero-shot protocol. Each VLM receives the rendered image and a fixed prompt that explains the circle/arrow/caption semantics, states the three validity conditions (correct, consistent, sufficient), and asks for a single-word RIGHT/WRONG answer; the identical prompt is used for open and closed models.

Pilot error profile. SmolVLM-500M makes 415 test errors: 122 false accepts and 293 false rejects. The weakest slices are arrow-between-destinations (83.1% correct rejection), wrong destination (88.0%), wrong magnitude (89.9%), and near-miss circle (91.7%)—exactly the geometrically subtle modes, which motivates H3.